Week 5: Introductory cybercrime and Forensics Lab

Before the analysis for proper folder arrangements and presentations, follow the following steps:

1. Create a folder with the Case number
2. Inside the new folder (case folder) create the following folder
    a. Image folder
    b. Autopsy folder
    c. Report Folder
    d. Document folder (the document should contain similar name as the case number)
    e. Hash folder
3. These folders should contain the content based on their name

image analysis

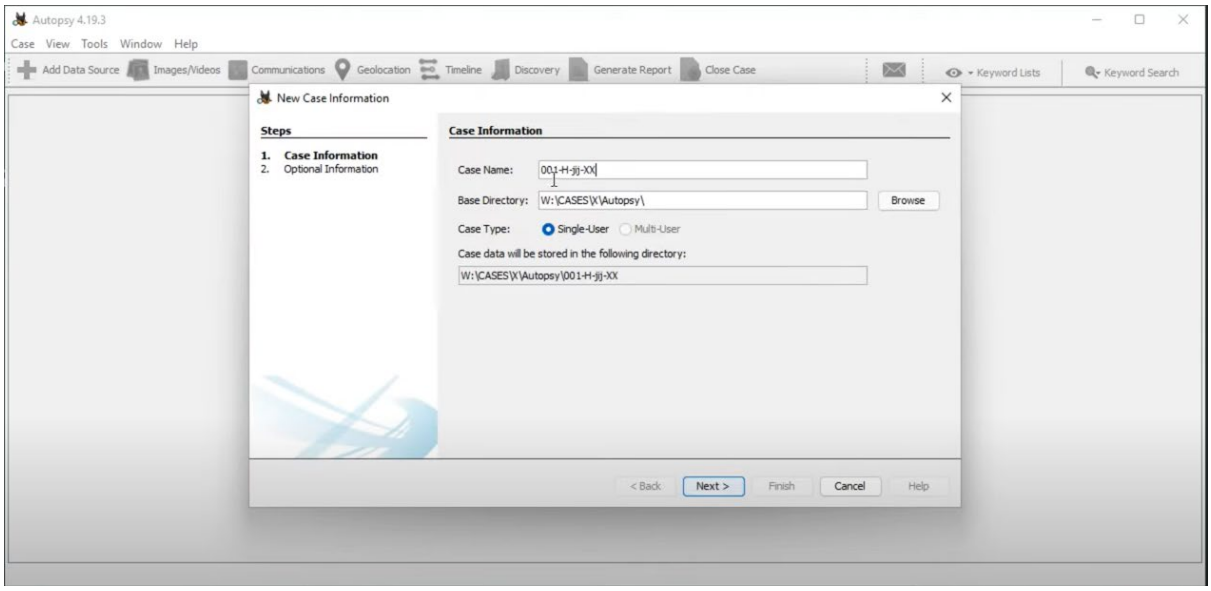
- Examine evidence(s) using autopsy
- Dig for evidence
- Examine the data carving
- Filtering,
- Searching,
- Report the evidence.

In this investigation, you will be working with *Mantooth*, *Washer* and *Thumbdrive* images. Add all the three evidences to Autopsy.

You do not have to finish all the steps, do as much as you can, and leave the rest to be completed on your spare time.

Sample of the starting point





Task

1. What are the file systems, and the volume serial numbers used in the Mantooth, Washer and Thumbdrive partitions? (Hint: Use the explore tab and look in properties window as well).
2. List all the users of the Mantooth drive.

3. How many files are there in the case from the three evidences?

4. How many files are there in each evidence, separately for Mantooth, Washer and Thumbdrive images?

5. How many files of category Microsoft World 2003 documents are there, in all evidences? How many are deleted? What is the content of the deleted file written by *confidential business letter.doc*?

6. in the file category, locate the raster graphics files. How many raster graphic files are there? Examine the content of all the files *minepart *.jpeg* and *minepart *.gif*. Are there any files that have contents related to banking cards and cheques?

7. Now examine the EXIF properties (Hint: use the properties tab to display in the properties window) of the file *mine part 19.jpeg, snuggle_kitty.jpg* and *harddrive.jpg*. Record the model of the cameras, the image models and the image date times. This is an important information to be used in the court of law during the cross-examinations.

Data carving

1. Examine the evidence and the files embedded into the image. Some files are encrypted, and some are deleted. Inspect the file status in the overview tab. Indicate how many files are encrypted and how many are deleted. How many data carved files are there?

2. Examine the encrypted files and bad extensions. How many deleted encrypted files? Can you view their content? How many files of bad extensions are there? Consider, for instance, the files starting with the name *20070*.gif* declared as bad extensions. Can you find the correct extensions? (Hint: use the file content pane with the tab HEX to deduce the correct extensions for some of the files. It should be written at the beginning of the file). Find who the owner of the files is (Hint: window of the properties). |why do you think, the suspect changed the extension of the files. (Anti-forensics)

3. Select the e-mail tab, expand the email status, e-mail archives and navigate to the folder *outlook.pst\Personal Folders\delete*d Items. View the attachments. View the file attached to the letter in the message sub container of the Email container.

4. Do further examinations on the *send items* in *outlook.pst* and check for the attachments. What is the date of birth of Tomas Book? Who is his family doctor? When did he visit his doctor? What is the address of the clinic? What medication he has been prescribed? What is the content of the file called password, in the email archives, *send items.dbx*? Who are the two persons exchanging emails?

Filtering

1. Filters can be applied to filter files by their metadata to extract particular evidence For instance, filter all files created after certain time or date.

2. Try to display only the graphic files. (Hint: Set the filter option to graphic files from the filter drop down menu).

3. Try again to display only Microsoft office files. How many Microsoft office files are there? Locate the *ATM_THEFTS1.ppt* and check it out in the file content. Open it in PowerPoint and examine it content.

4. Locate and export the file "*How to Steak credit numbers.doc*". This file is encrypted.

5. Create a new filter with the rules to display only *JPEG* files and/or word *2003 documents*